

SCHOOL MANAGEMENT SYSTEM (SchoolMS)

PROJECT PROPOSAL DOCUMENT – ADVANCED VERSION

1. PROPOSAL SUMMARY

We propose to develop an **Advanced School Management System (SchoolMS)** with a professional website, smart automation, face recognition attendance, online learning module (LMS), library and transport management, and comprehensive analytics. Ideal for **large schools and multi-branch institutions (1000-3000 students)** seeking complete digital transformation.

Ideal For: Large schools, multi-branch chains, residential schools, group of institutions (1000-3000 students)

2. OBJECTIVES

- Fully digitize all academic and administrative workflows
- Automate attendance, fee, exam, and communication processes
- Implement AI-based face recognition for student and staff attendance
- Provide an integrated Learning Management System (LMS)
- Enable real-time multi-branch monitoring and reporting

3. SCOPE OF WORK

3.1 Website Development (Advanced)

- Premium school website with modern animations
- Online admission portal with document upload
- Virtual campus tour section
- Student and parent login portal
- Event calendar, news and achievements section
- School app download links integration

Cost: Rs. 40,000 - Rs. 60,000

3.2 Student Management (Advanced)

- Complete student lifecycle management (Inquiry to Alumni)
- Digital admission form with document verification
- Student ID card with QR code
- Promotion and TC management

- Health and medical record maintenance
- Co-curricular activities and achievements tracking

Cost: Rs. 55,000 - Rs. 80,000

3.3 Attendance - Face Recognition System

- AI face recognition-based student auto attendance
- Teacher and staff face recognition check-in
- Real-time attendance dashboard
- Instant parent alert on absence via WhatsApp/SMS
- Monthly and yearly attendance analytics

Cost: Rs. 80,000 - Rs. 1,20,000

3.4 Fee Management (Advanced)

- Fully automated fee cycle management
- Online payment gateway with auto receipt
- Multiple fee heads (tuition/transport/hostel)
- Scholarship and concession module
- Fee defaulter tracking and auto reminders
- GST-compliant invoicing and ledger

Cost: Rs. 45,000 - Rs. 65,000

3.5 Learning Management System (LMS)

- Chapter-wise study material upload (PDF/Video)
- Online assignment submission by students
- Teacher grading and feedback portal
- Live class scheduling (Zoom/Meet integration)
- Student progress tracking per subject

Cost: Rs. 60,000 - Rs. 90,000

3.6 Exam, Result & Report Card

- Online exam scheduling and conduction
- OMR-style objective test module
- Auto result calculation with CGPA/grades
- Customizable report card templates
- Parent result sharing via portal and WhatsApp

Cost: Rs. 45,000 - Rs. 65,000

3.7 Library Management

- Book catalog with barcode/QR tracking
- Issue, return and renewal management
- Overdue fine calculation

- Student-wise library history

Cost: Rs. 25,000 - Rs. 38,000

3.8 Transport Management

- Route and vehicle management
- Student-wise bus allocation
- Driver and conductor profiles
- GPS live tracking integration (basic)

Cost: Rs. 28,000 - Rs. 42,000

3.9 Staff & HR Management

- Complete staff profiles with document storage
- Payroll with salary slip generation
- Leave and shift management
- Performance appraisal module

Cost: Rs. 40,000 - Rs. 60,000

3.10 Admin Dashboard (Advanced Analytics)

- Multi-branch real-time school dashboard
- Student academic performance analytics
- Fee collection and defaulter reports
- Staff attendance and HR insights
- Custom report export (PDF/Excel)

Cost: Rs. 38,000 - Rs. 55,000

4. ADDITIONAL FEATURES

4.1 WhatsApp + SMS + Email Notifications

- Daily attendance alert to parents
- Exam results and fee receipts on WhatsApp
- School circulars and event announcements
- Automated birthday and achievement wishes

Cost: Rs. 20,000 - Rs. 30,000

4.2 Parent & Student Portal (Advanced)

- Full-featured parent login dashboard
- Live attendance, results, homework, fee access
- Online leave application by parents
- Two-way teacher-parent messaging

Cost: Rs. 22,000 - Rs. 32,000

4.3 Hostel Management

- Room and bed allocation management
- Hostel fee tracking
- Visitor log management
- Warden and housekeeping staff records

Cost: Rs. 20,000 - Rs. 32,000

4.4 Role-Based Access & Security

- Multi-level roles (Admin/Principal/Teacher/Parent/Student)
- Secure JWT authentication

Cost: Rs. 15,000 - Rs. 22,000

4.5 Cloud Backup & Audit Logs

- Daily automated cloud backup
- Full system activity and access logs

Cost: Rs. 12,000 - Rs. 18,000

5. TECHNOLOGY STACK (MERN + AI)

- **Frontend:** React.js
- **Backend:** Node.js (Express.js)
- **Database:** MongoDB
- **AI Integration:** OpenCV / AWS Rekognition (Face Recognition)
- **Live Class:** Zoom / Google Meet API
- **Payment:** Razorpay / PayU
- **Hosting:** AWS Cloud

6. TOTAL PROJECT COSTING

Development Cost (One-time)	Rs. 5,00,000 - Rs. 7,50,000 (One-time)
Infrastructure	Hosting: Rs. 4,000 - Rs. 10,000/month
Infrastructure	Domain + SSL: Rs. 2,000/year
Infrastructure	WhatsApp Business API: Rs. 2,000 - Rs. 5,000/month
Maintenance Cost	Rs. 28,000 - Rs. 40,000/month/month

7. PROJECT TIMELINE

Phase	Duration
-------	----------

Planning	1 Week
Development	8-10 Weeks
Testing	1-2 Weeks
Deployment	1 Week
Total Duration	10-13 Weeks

8. LIMITATIONS

- No dedicated mobile app (can be added later)
- No AI-based academic performance prediction
- No full ERP integration
- Limited IoT/smart classroom integration

9. FUTURE UPGRADES

- Mobile app for students, parents and teachers
- AI-based academic performance prediction
- Full school ERP integration
- IoT smart classroom and bell system integration

10. SUPPORT AND WARRANTY

- 1 month free support
- Bug fixing included
- Staff training provided

11. TERMS AND CONDITIONS

- 50% advance payment
- 50% after completion
- Additional features charged separately

12. CONCLUSION

This Advanced SchoolIMS provides a **powerful, scalable, and highly automated platform** — combining face recognition attendance, LMS, online exams, transport, library, WhatsApp communication, and multi-branch analytics, making it ideal for schools committed to complete digital transformation.

13. CONTACT DETAILS

Company Name: Kartsho Digital
Parent Company: A Venture of Kartsho Enterprises
Phone: +91 9453134901 / +91 9528660578
Email: kartshoenterprises@gmail.com
Website: digital.kartsho.com
Address: Godhoopatti, Patti, Pratapgarh, Uttar Pradesh - 230134